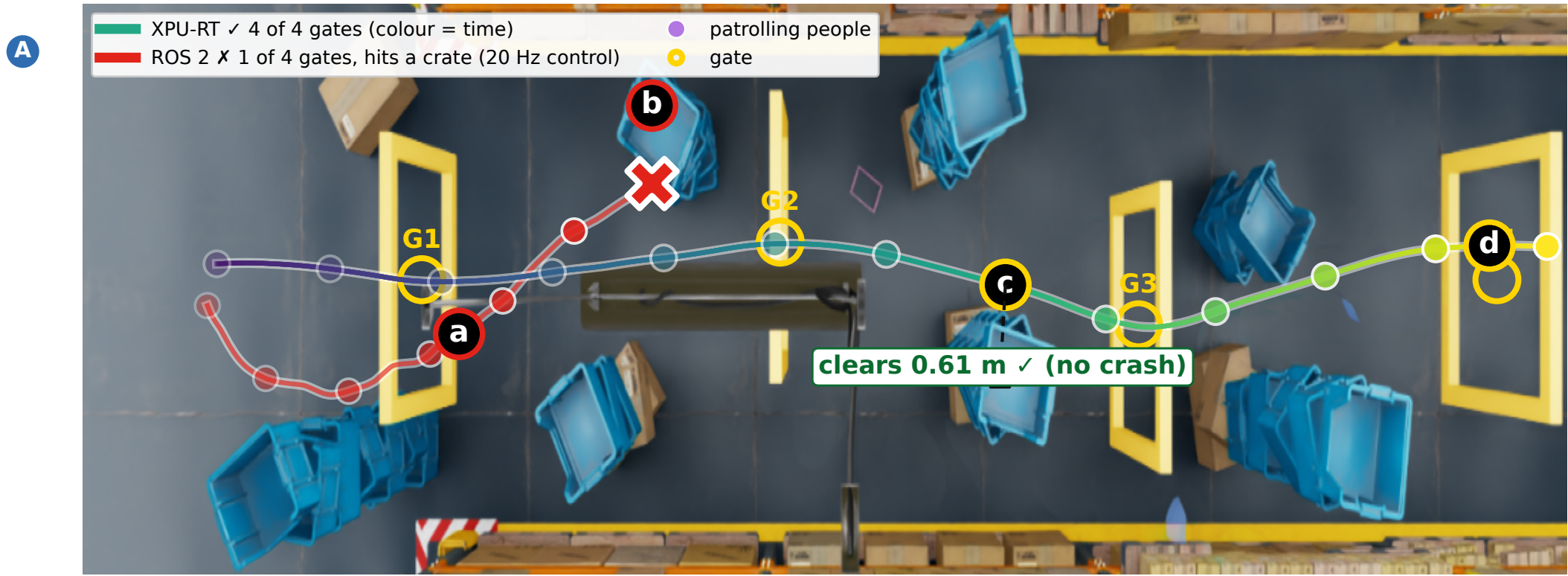
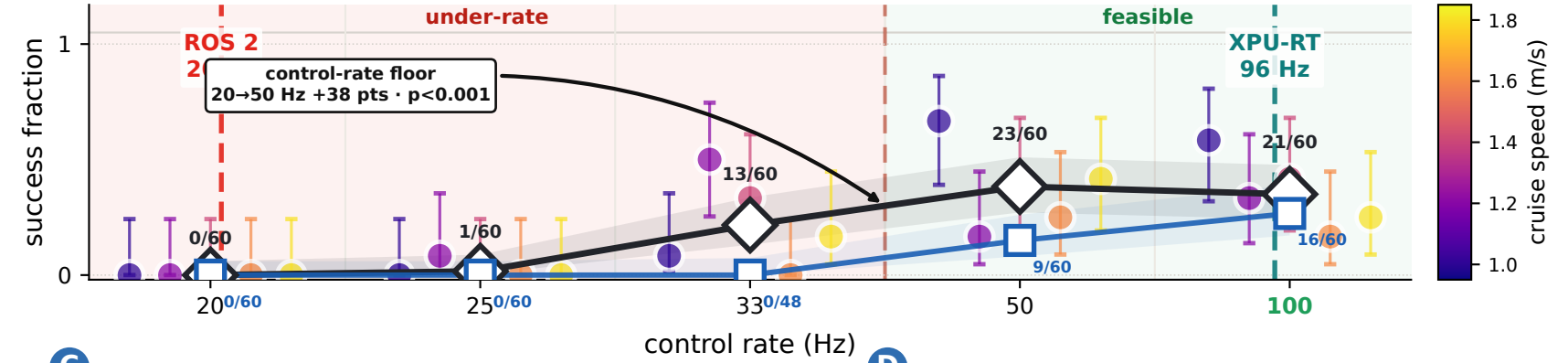


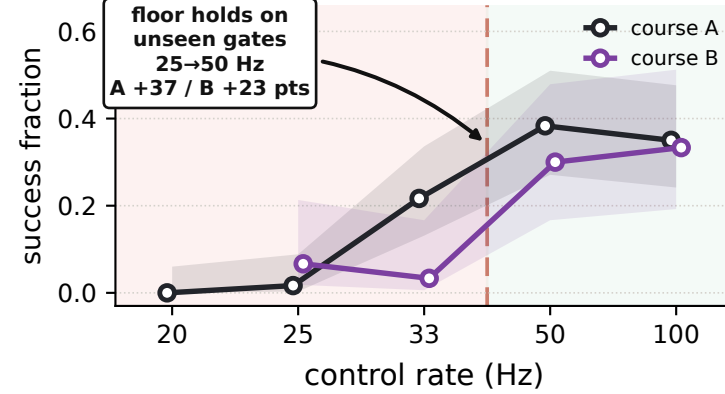
Warehouse gate-showdown — same scene and drone, gain, at 1.4 m/s
XPU-RT · CP-SAT (96 Hz control) clears all 4 gates
ROS 2 · serial YOLO, control in the goal callback (20 Hz control) clears G1 → hits a crate
one seed of this scene (seed 1000); over its 12 seeds XPU-RT completes 3 and ROS 2 none



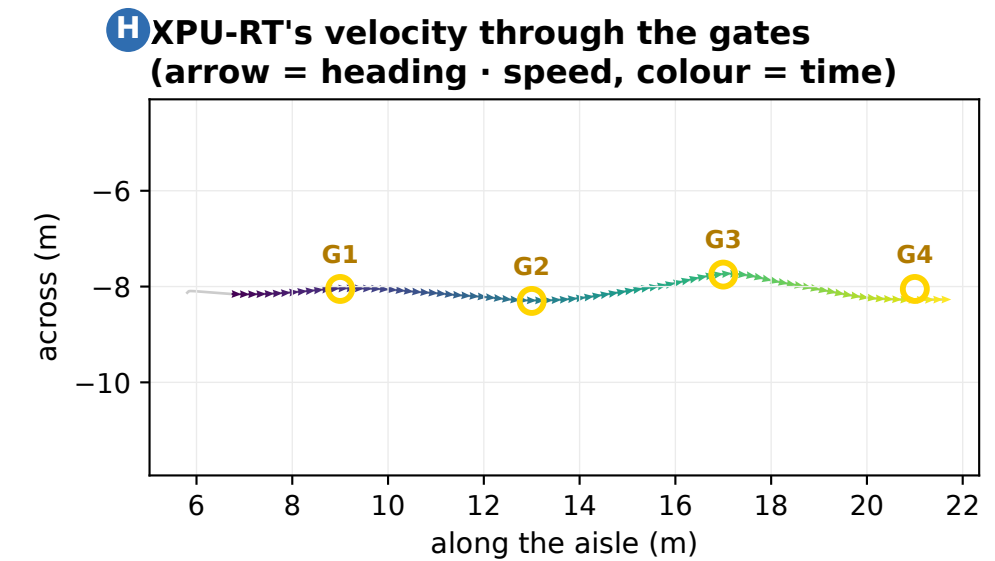
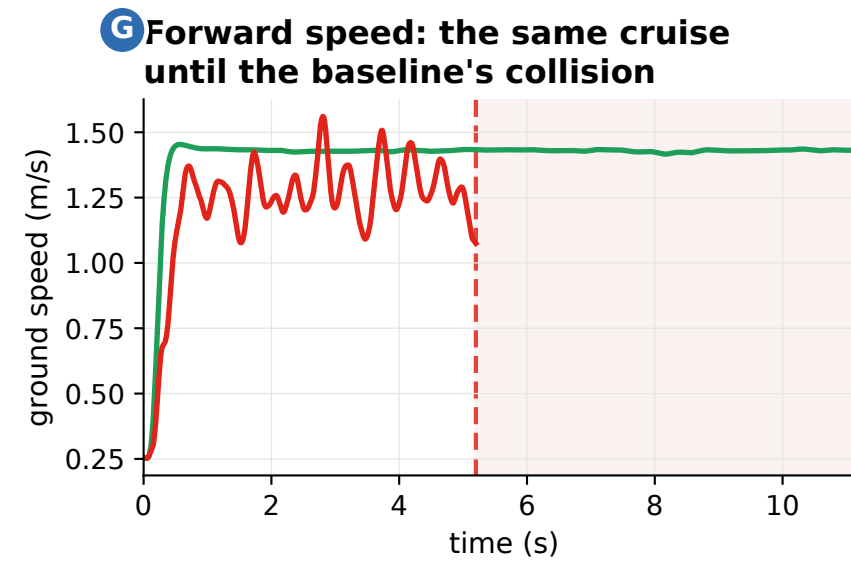
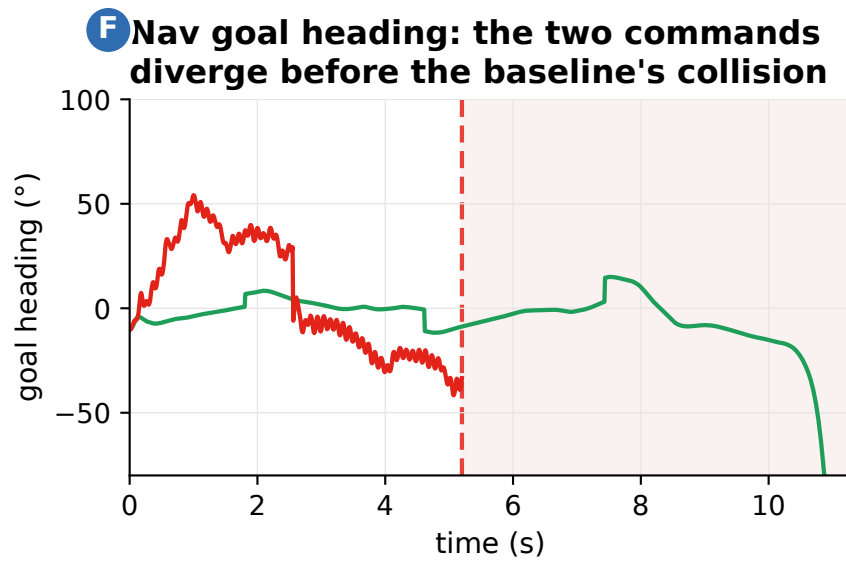
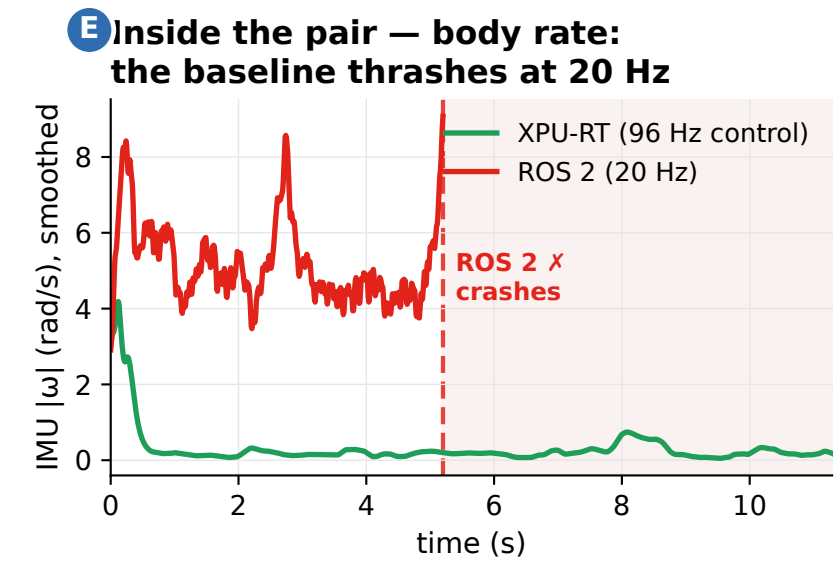
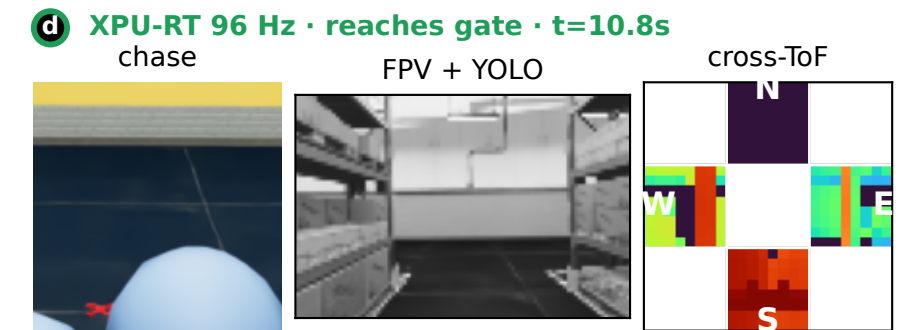
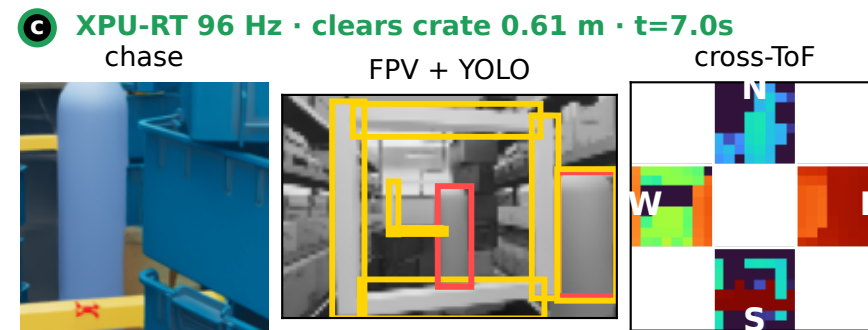
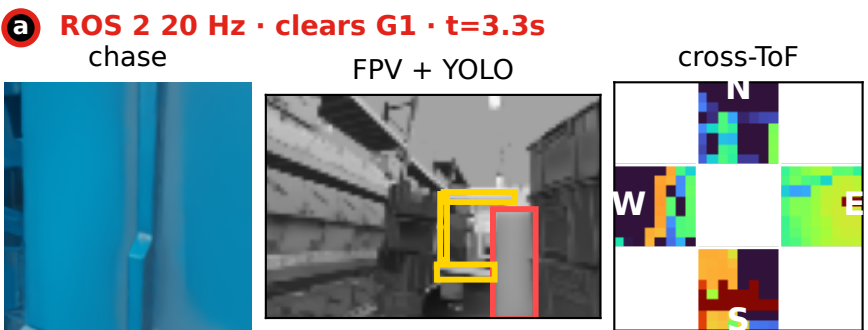
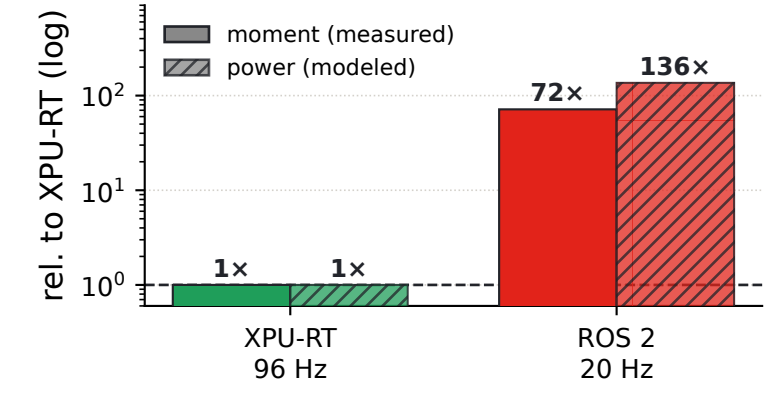
B Control-rate envelope (rate injected, 300 flights): success against control rate — ROS 2's 20 Hz and XPU-RT's 96 Hz marked



C Generalization: unseen gates (course B)



D Mechanism: the baseline thrashes (n=6 flights/arm)



I Onboard K1 schedule, measured at the 45 Hz camera — XPU-RT · CP-SAT places the kernels over all 8 harts: camera→control 57 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, control on 8 harts, nav on 8 harts
ROS 2 (4 processes · single executor · chained control · unpinned): camera→control 266 ms, control every 49.6 ms, 401/401 frames late · YOLO on 1 hart, nav on 1 hart, control on 1 hart

